TSA Mock 2

Problem Solving

The hardest past TSA Problem Solving questions not in Mock 1, any type, from all 16 papers (2008–2023)

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Every question is a real past TSA question. The year and number in brackets show where it comes from, e.g. **7 (2019 Q32)**.

What is in this paper

The 25 Problem Solving questions judged hardest of all those not already in Mock 1. Type does not matter here, so questions that do not fit neatly into one type are included too.

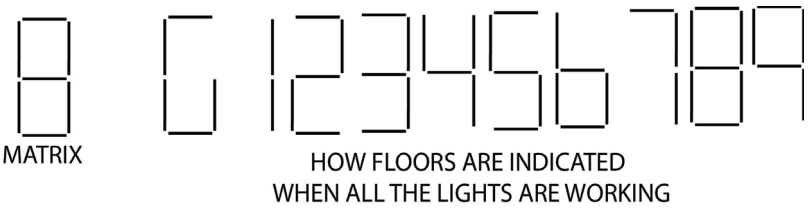
How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to 25 questions: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes). These are hard questions, so expect to run short of time.

Score: _____ / 25

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1 (2014 Q44) Allan wishes to visit a client on floor 8 of a building with 10 floors (including the ground floor). He enters the lift on the ground floor. His client has explained by email that the floors are indicated by an illuminated matrix but there is a notice stating that two of the lights are not working.



MATRIX

HOW FLOORS ARE INDICATED
WHEN ALL THE LIGHTS ARE WORKING

Allan enters the lift and pushes the button. Just then he starts to sneeze, distracting his attention from the display. The lift leaves the ground floor, travels, then stops. He glances at the display and concludes he could be on floor 8 so he gets out. In fact he is on the wrong floor.

On how many of the floors could Allan have got out by mistake?

A 5

B 6

C 7

D 8

E 9

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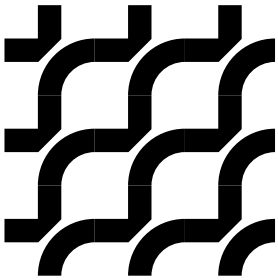
2 (2008 Q26)

The design of a tile for a bathroom wall is shown below:

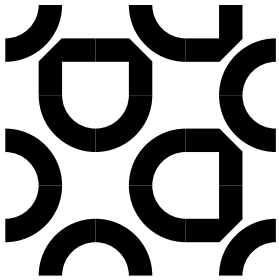


Which of the following patterns may **NOT** be made using only a number of identical tiles as shown?

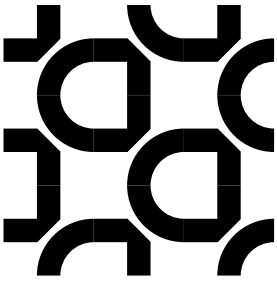
A



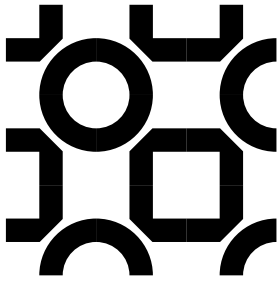
B



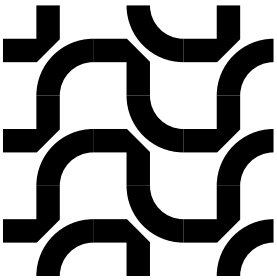
C



D



E



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3 (2021 Q30) I am planning a party next weekend. I have invited 100 people. Last week I took advantage of an offer at my local supermarket where I was able to get 10% off the total price of party food if I spent more than $75. I paid $108 and I estimate that the food that I bought will be enough for 60 people. I've now had replies to the invitations and I know that there will be 80 people at the party. This afternoon I will buy the remaining food that I need.

What is the total price of the extra food that I need?

A $32.40

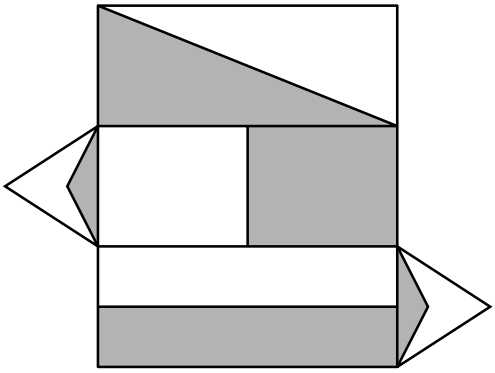
B $36.00

C $40.00

D $72.00

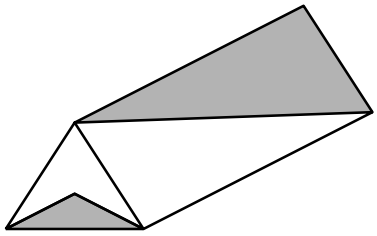
E $80.00

4 (2011 Q42) The piece of card shown below is folded to make a box with a triangular cross-section.

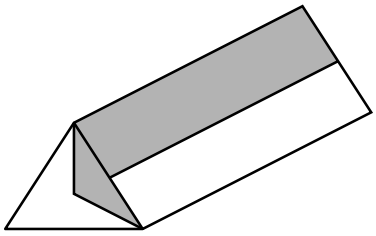


Which one of the views below can **not** be of this box?

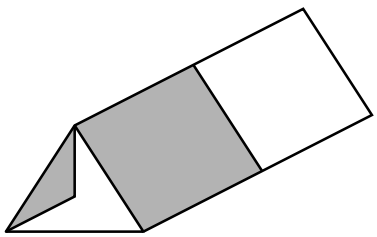
A



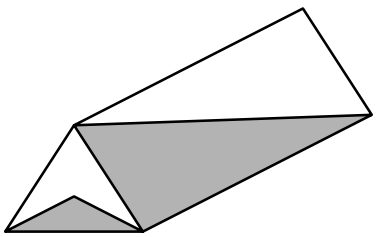
B



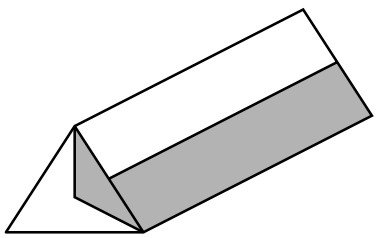
C



D



E



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5 (2023 Q8) As I waited for a train yesterday I was puzzled when I noticed the digital clock on the platform change from



to



but then a minute later it changed to



And then, after another minute, it changed to



I realised that the last digit was not displaying properly.

What was wrong with the last digit?

A One element was permanently on and one element was permanently off.

B One element was permanently on and two elements were permanently off.

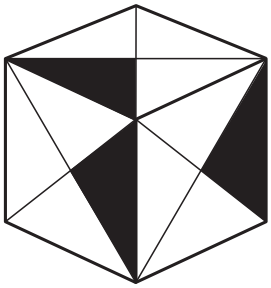
C Two elements were permanently on and one element was permanently off.

D Two elements were permanently on and two elements were permanently off.

E Three elements were permanently on and one element was permanently off.

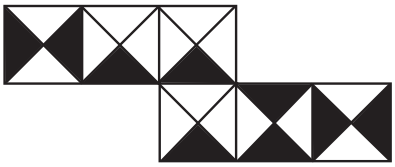
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6 Here is a diagram of a cube.
(2018 Q44)

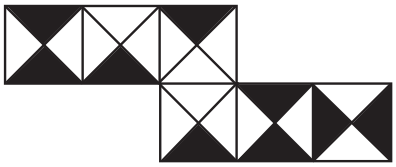


Which one of the following nets **could** be folded to make this cube?

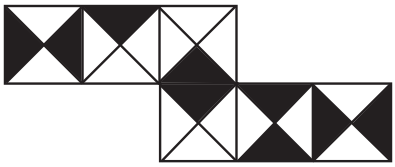
1



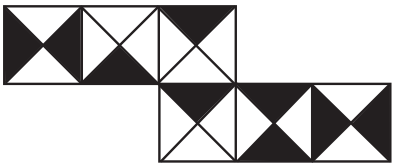
2



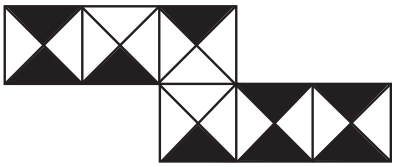
3



4



5



A 1

B 2

C 3

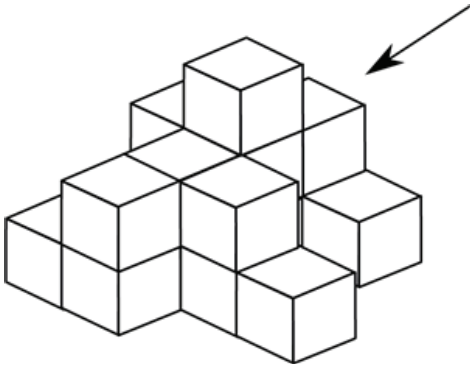
D 4

E 5

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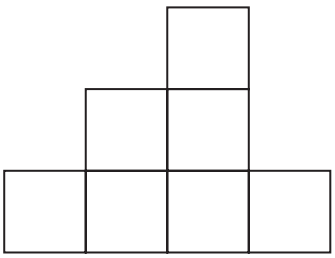
7 (2016 Q20) One of the exhibits in the Interactive Art Exhibition at the Hanson Gallery consists of 16 identical free-standing cubes. Visitors to the exhibition are invited to handle these cubes and rearrange them to create their own work of art.

This is the arrangement that I have made:

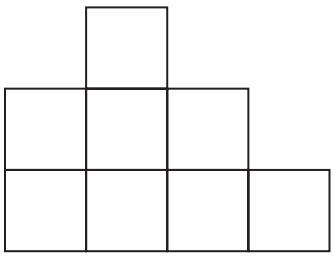


What is the side view of my arrangement from the direction indicated by the arrow?

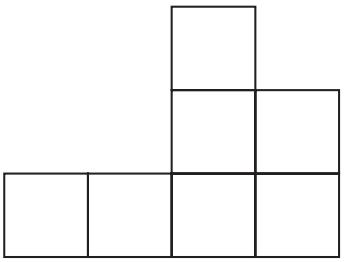
A



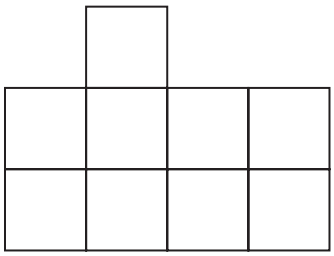
B



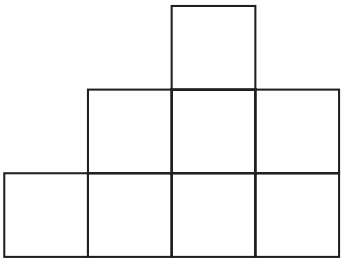
C



D

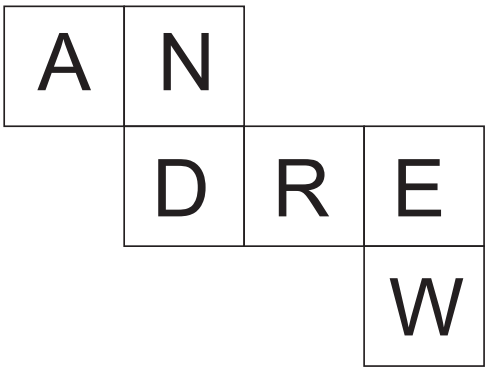


E



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8 (2018 Q20) The diagram below shows the net of a cube.



The following diagrams show five suggested views of the cube once it has been assembled.



1 2 3 4 5

Which two of the above are possible views of the cube?

A 1 and 2

B 2 and 3

C 3 and 4

D 1 and 4

E 4 and 5

9 (2013 Q6) On only 360 occasions during the twentieth century was it possible to write the date in the form DD/MM/YYYY using eight different digits. One example was 28th July 1956 (28/07/1956).

Eleven of Arthur's birthdays have been such dates (though not his date of birth), and in 1974 and 1983 the two digits that did not form part of the date made up his age at the time.

In which year was Arthur born?

A 1909

B 1916

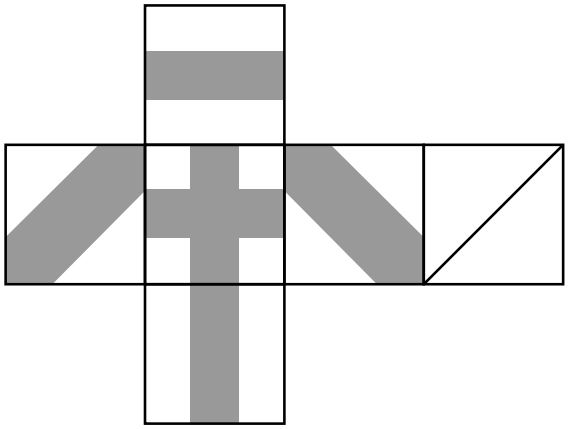
C 1918

D 1936

E 1938

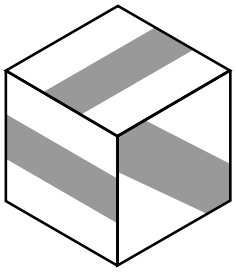
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10 The cut out below was folded together to make a cube.
(2008 Q14)

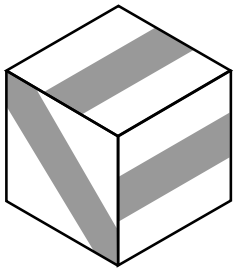


Which of the following cubes was formed from this cut-out?

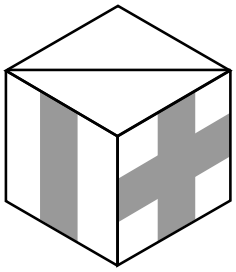
A



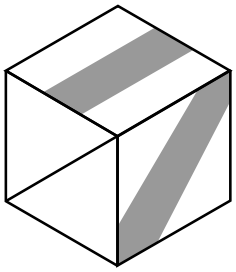
B



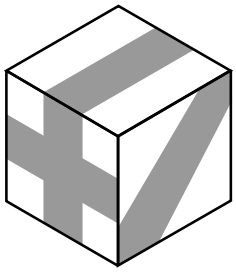
C



D



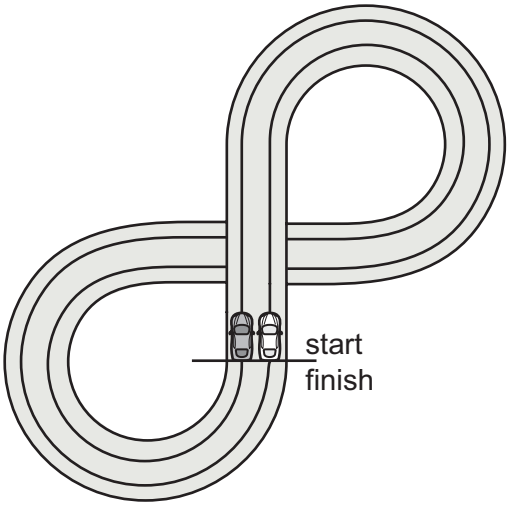
E



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11 The drawing shows a slot car track.
(2023 Q46)

A grey car and a white car start together at the same point and travel in the same direction. Because of the figure-of-eight arrangement of the track, each car travels exactly the same distance to complete one lap of the track. The grey car is 12% faster than the white car.



How many laps will the grey car have completed before it passes the white car for the first time?

A 8

B 9

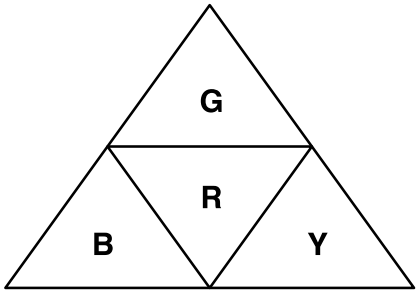
C 10

D 11

E 12

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12 Four triangles of card coloured on one side are joined as shown.
(2009 Q50)

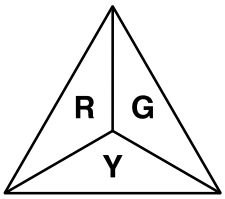


G = Green
R = Red
B = Blue
Y = Yellow

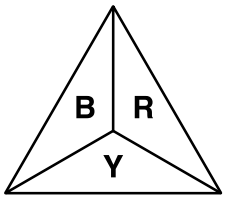
The joined triangles are then folded and joined to make a pyramid.

Which of the following could **not** be a view from above?

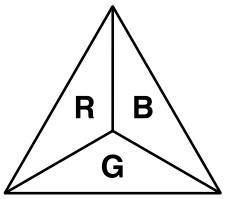
A



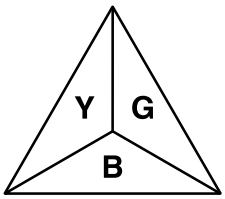
B



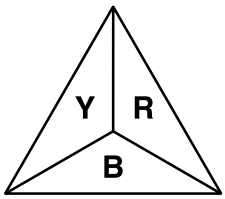
C



D

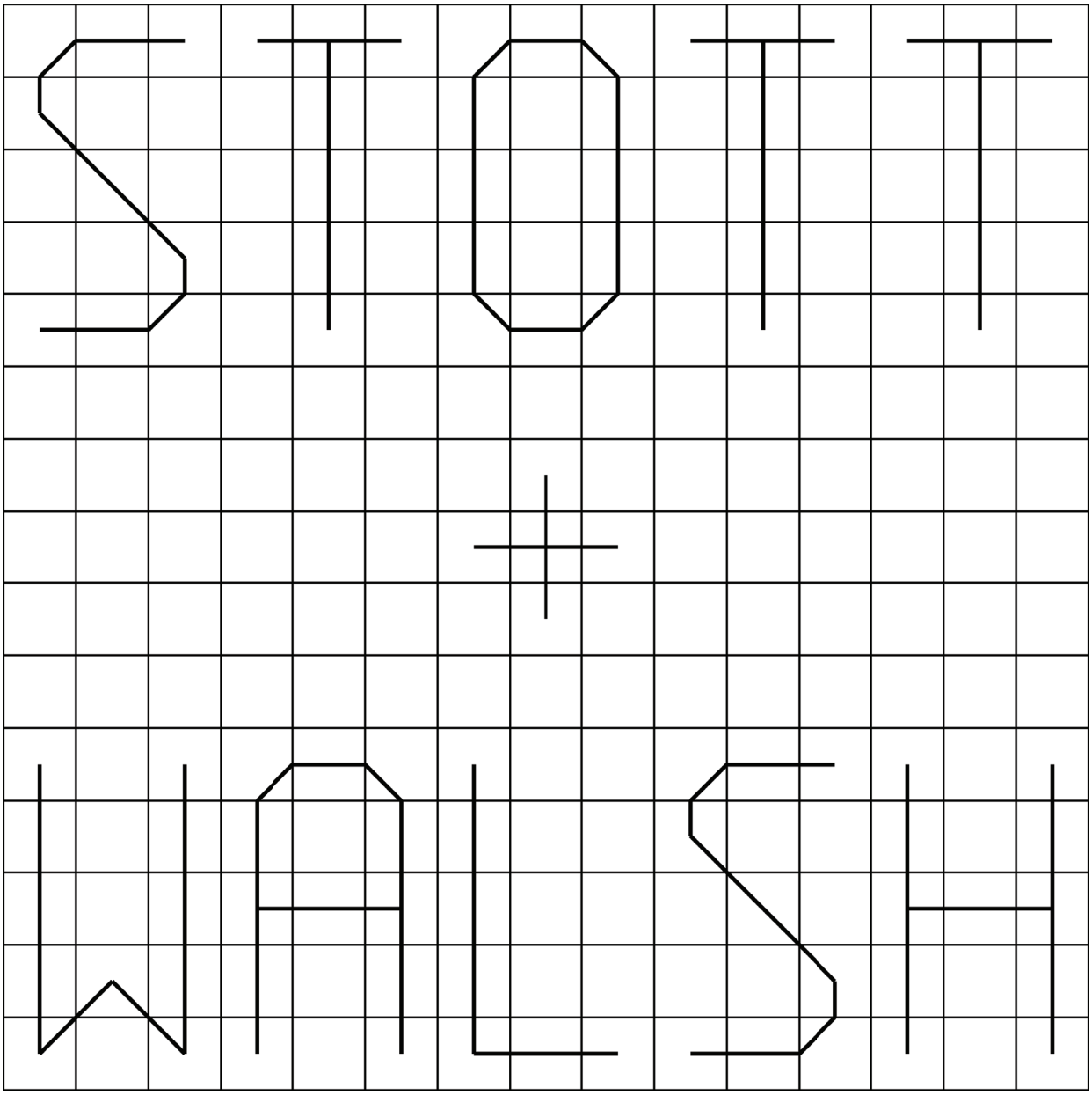


E

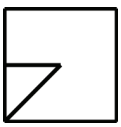


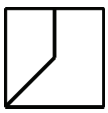
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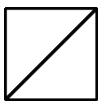
13 The floor of the reception area of the head office of Stott and Walsh is tiled as shown below:
(2016 Q14)



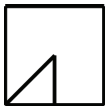
Which one of these tiles could **not** be used to replace a damaged tile on this floor?

A 

B 

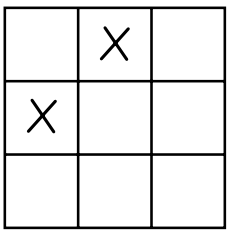
C 

D 

E 

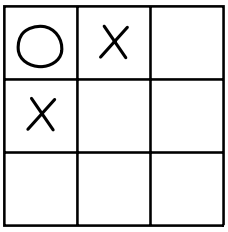
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14 (2011 Q31) In a modified form of a popular game, each of two players at any turn may put either 'O' or 'X' as she chooses in any blank square of a 3 by 3 grid. She wins if she thereby completes a line of three identical symbols. A player is faced with this situation:

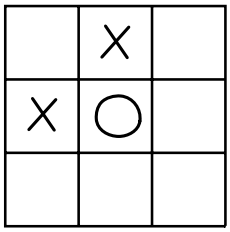


Which one of the following moves ensures that she can win at her next move?

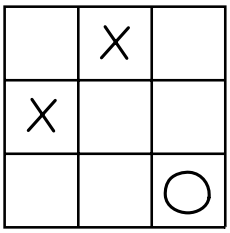
A



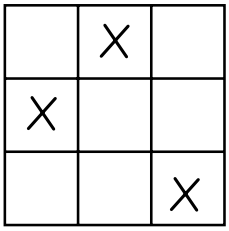
B



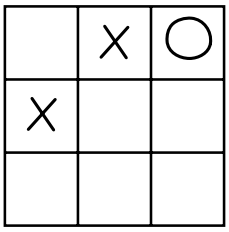
C



D

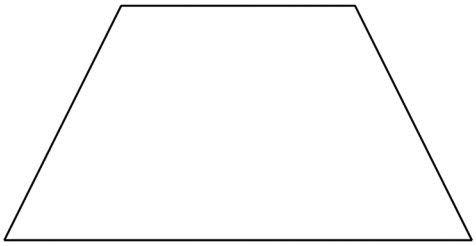


E



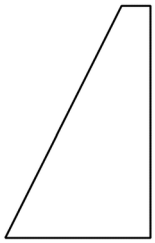
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15 The trapezium below may be cut into four identical pieces.
(2012 Q32)

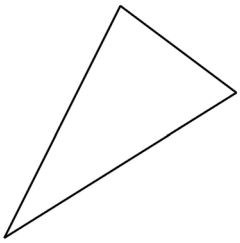


Which of the following pieces can be used four times (flips and rotations are allowed) to make up the trapezium?

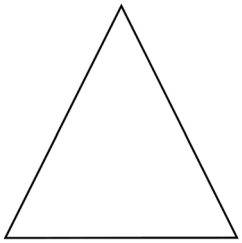
A



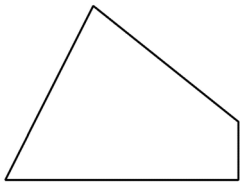
B



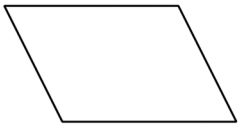
C



D

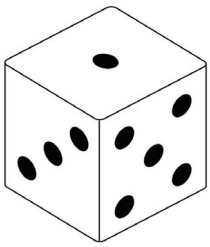


E



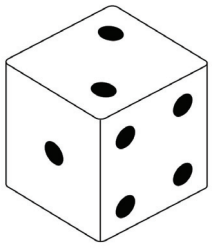
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16 (2015 Q44) The die shown below is a conventional die, i.e. the total number of spots on any two opposite faces is always seven.

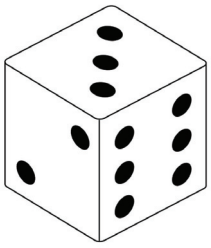


Which one of the following could **not** be the appearance of the die shown above after a subsequent throw?

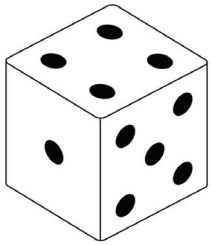
A



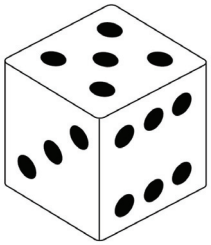
B



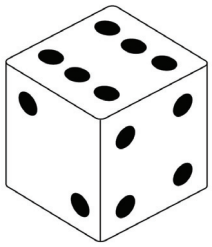
C



D

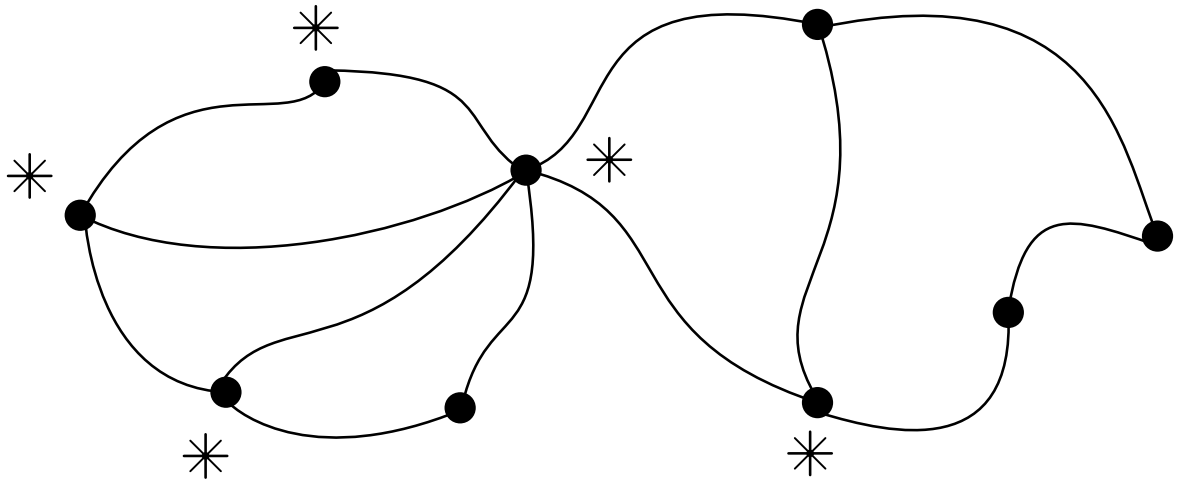


E



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17 (2010 Q44) My local park has a network of paths linking various locations, as shown below. The stars on the map represent five of the park's landmarks: the fountain, lake, bandstand, statue and bowling green.



When I walk round the park, I often vary my routes and I do not always take the shortest route from one place to another. I have noticed that when I go from the statue to the fountain, I always pass the bandstand, but that going from the lake to the bowling green, I never pass the fountain unless I pass the bandstand twice.

How many of the stars on the diagram can be identified from the given information?

A 0
B 1
C 2
D 3
E 5

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18 (2020 Q32) Bert has agreed to make 14 decorations for his daughter's birthday. Each decoration will require two gold ribbons, one 12 cm long and the other 7 cm long.

The table shows the lengths and prices of available gold ribbon.

gold ribbon price list	
gold ribbon length	price
30 cm	£4.00
50 cm	£5.00
60 cm	£6.00

The ribbon can be easily cut to size for the decorations, but short pieces of ribbon cannot be joined to make longer lengths.

What is the least amount that Bert will need to spend on gold ribbon for the 14 decorations?

A £26.00

B £27.00

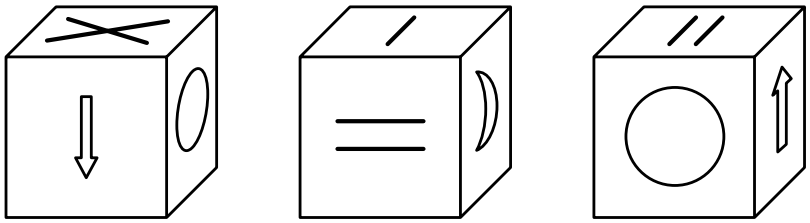
C £28.00

D £29.00

E £30.00

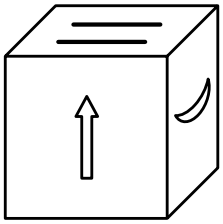
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19 Three views of the same die with unusual symbols are shown below:
(2010 Q38)

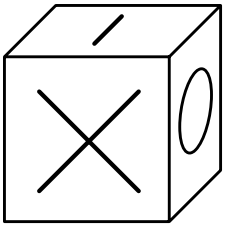


Which one of the following dice is **not** identical to the die above?

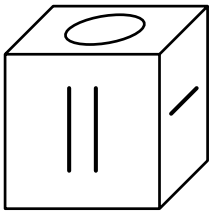
A



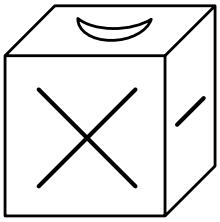
B



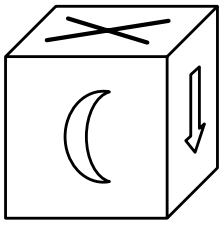
C



D

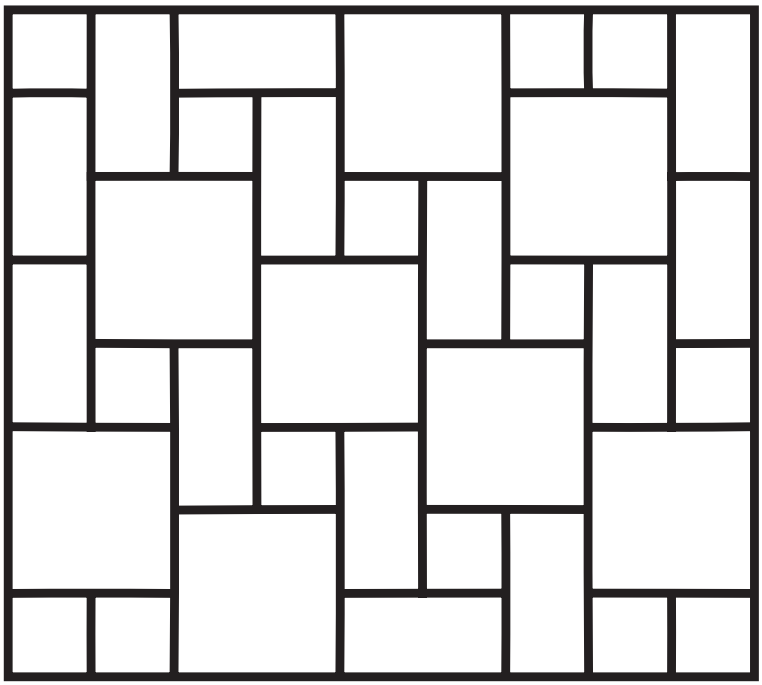


E



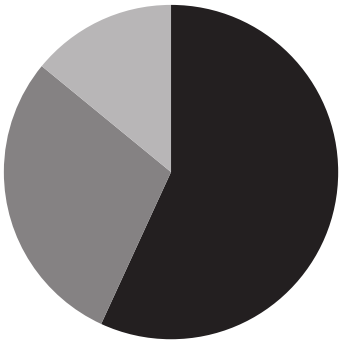
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20 The block paved area shown below is laid using three different-sized blocks.
(2023 Q20)

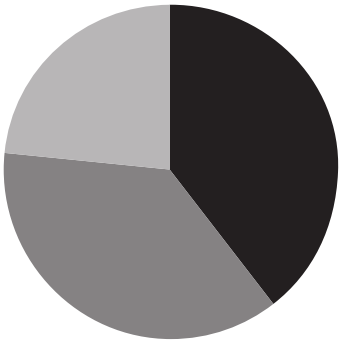


Which one of the following charts shows the relative comparison of the number of each size of block used in this paved area?

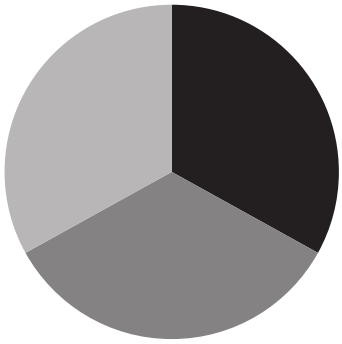
A



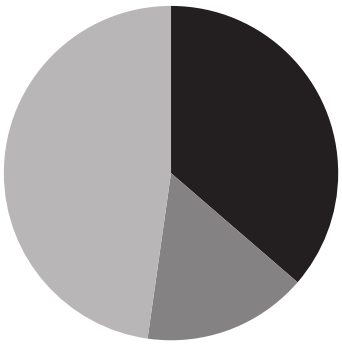
B



C



D

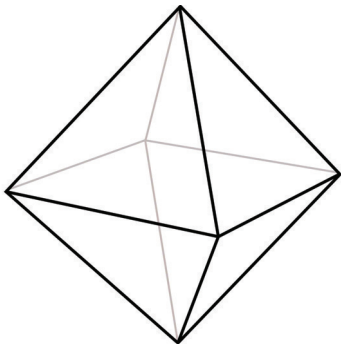


E



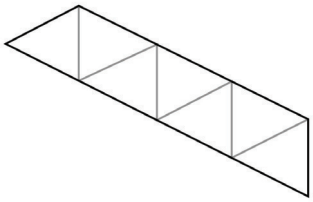
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21 (2014 Q32) A regular octahedron is a solid composed of eight equilateral triangles, as shown in the image below:

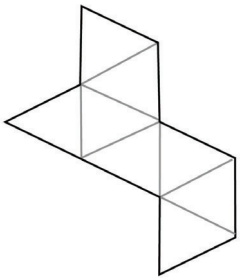


Which one of the following paper shapes will fold into an octahedron?

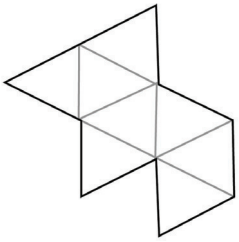
A



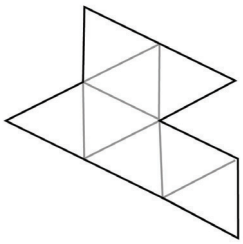
B



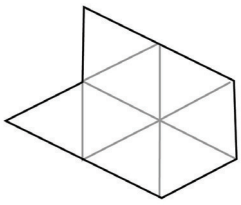
C



D

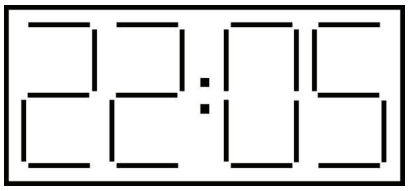


E



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22 (2015 Q38) I stayed in a hotel last night. My room was at the front of the building. At one point I looked out of the window and saw:

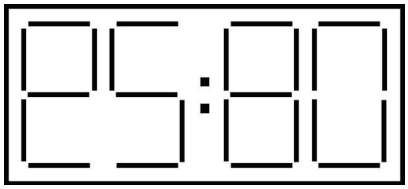


on an illuminated digital clock on the other side of the road. I took the opportunity to reset my watch, which had stopped sometime earlier.

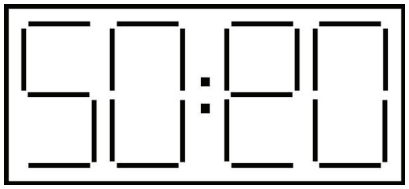
This morning, when I looked out, I realised that what I could see across the road was a reflection in the building opposite of a clock outside the hotel. According to my watch (which I now knew I had reset incorrectly last night) the time was supposedly nine minutes past ten.

What did I see across the road when I looked out this morning?

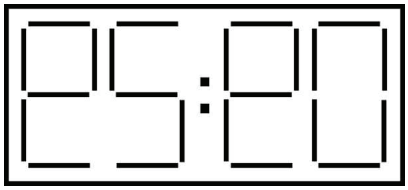
A



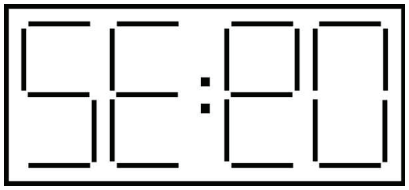
B



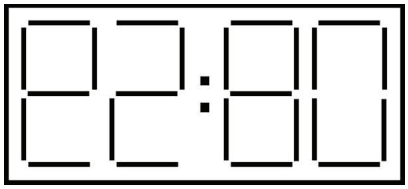
C



D

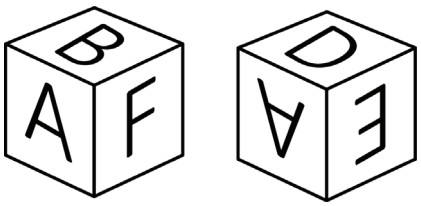


E



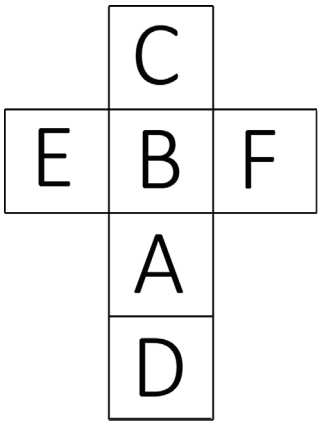
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23 (2016 Q8) A cube is labelled with six letters and, when viewed in two positions, looks like this:

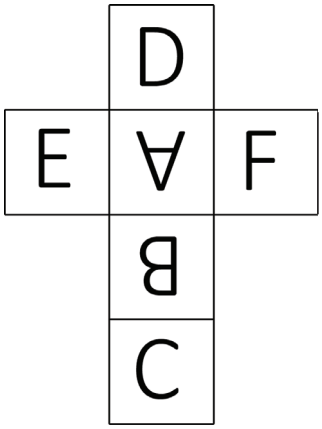


Which one of the following nets could be folded to make the above cube?

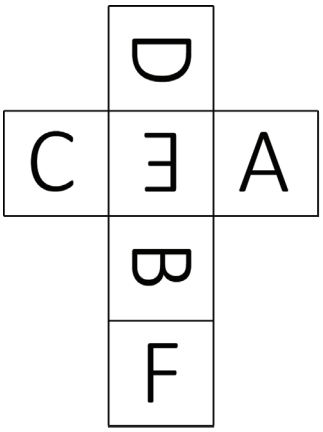
A



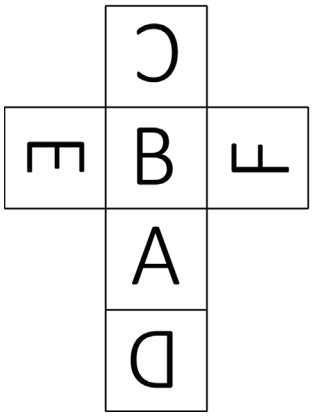
B



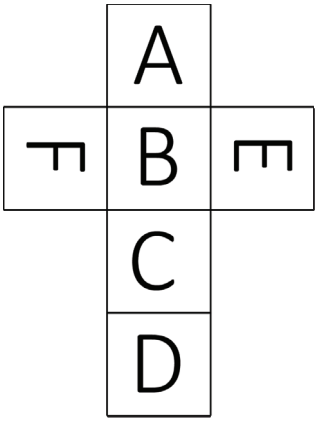
C



D

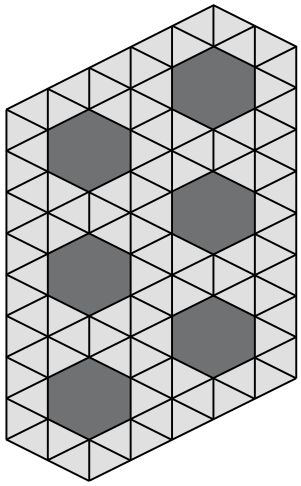


E



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24 (2009 Q44) The pattern below shows part of the tiling of a large floor in a community centre. It is made up of two types of tile: a hexagon and a triangle.



For a sufficiently large expanse of flooring, approximately what proportion of the two types of tile will be required?

A 1 hexagon to 6 triangles

B 1 hexagon to 9 triangles

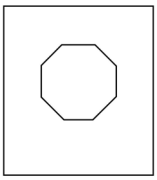
C 1 hexagon to 12 triangles

D 1 hexagon to 15 triangles

E 1 hexagon to 18 triangles

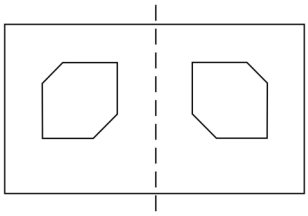
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25 (2012 Q26) A sheet of paper has two holes cut from it. When it is folded in half the hole shown in the diagram below is seen.

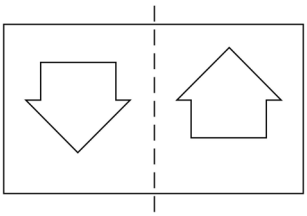


Which one of the following could **not** be the unfolded sheet of paper? (The dotted line indicates where it was folded.)

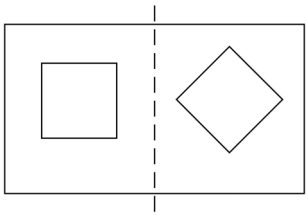
A



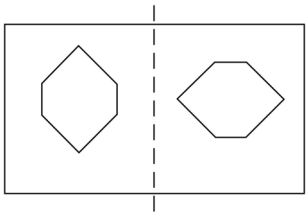
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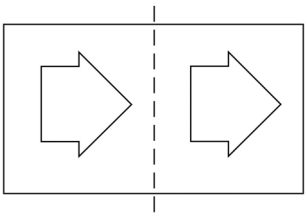
C



D



E



END OF TEST

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